OP-OG : 150A

Balances:

Mass

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho v) = 0 \qquad \nabla \cdot v = 0 \ \text{(Incompressibility)}$$

Mass (Macro)

$$\frac{\partial}{\partial t}\int_{CV} \rho \, dV = -\iint_{CS_1} \rho (v \cdot n)\, dA - \iint_{CS_2} \rho (v \cdot n)\, dA$$

$$\frac{dm}{dt} = \dot{m}_{in} - \dot{m}_{out}$$

$\beta = \frac{4}{3}$ (pipe)

$\beta = 1$ (plug)

$$B = \frac{\langle v^2 \rangle}{\langle v \rangle^2}$$

(Macro) Momenta

$$\Sigma F = \frac{\partial}{\partial t}\int_{CV} v \rho \, dV + \iint_{CS} v \rho (v \cdot n)\, dA$$

$$\iint_{CS} v \rho (v \cdot n)\, dA = \sum_{CS} \rho \langle v^2 \rangle A n_\perp = \sum_{CS} \dot{m} \langle v \rangle \beta n_\perp$$

Energy

$$\dot{m}\Delta H = \dot{Q} + W_s \quad \text{(Steady state Open system)}$$

$$\frac{\partial U}{\partial t} = \dot{Q} + W_s \quad \text{(Unsteady Closed system)}$$

$$y_1 + \frac{v_1^2}{2g} + \frac{P_1}{\rho g} = y_2 + \frac{v_2^2}{2g} + \frac{P_2}{\rho g} + h_L \quad \text{(Bernoulli)}$$

$$\frac{P_A}{\rho} + \frac{1}{2}\alpha_A \langle v_A \rangle^2 + g h_A = \frac{P_B}{\rho} + \frac{1}{2}\alpha_B \langle v_A \rangle^2 + g h_B + \hat{E}_v + \hat{W}_s$$

$$\Delta P = P_2 - P_1 + \rho g \Delta h$$

$\langle v \rangle = \langle v_B \rangle \Rightarrow \alpha = 1$

$$\hat{E}_v = \frac{E_v}{\rho V_1 A_1} = h_L$$

$$\dot{Q} - \dot{W}_s = \frac{\partial}{\partial t}\int_{CV} e \rho \, dV + \iint_{CS} \left(e + \frac{P}{\rho}\right) \rho (v \cdot n)\, dA + \overbrace{\int (\tau^T \cdot n) v \, dA}^{\text{Viscous}}$$

Torque

$$\Sigma M = I \frac{d\omega}{dt} \qquad \underline{\underline{\tau}} = \tau^T \ (\tau_{xy} = \tau_{yx})$$

$$\nabla \times \underline{v} = 0 \ \text{(Irrotational flow)}$$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \quad ; \quad \omega = \frac{v}{r} \quad , \quad \underbrace{\tau = F_\tau R}_{\text{Frictional moment}} \ ; \quad \tau \cdot A \quad ; \quad P = F \cdot v = A\mu \frac{\partial v}{\partial y} v = \frac{A\mu v^2}{h}$$

$\tau_{yx} = \mu \frac{\partial v_2}{\partial y}$ (τ_{yx}: normal → Direction)

$$\langle v \rangle = \frac{1}{A}\iint_{CS} v \cdot n_\perp \, dA \qquad Q = \int v \cdot dA \ \text{(Volumetric flow rate)} \qquad \langle v \rangle = \frac{Q}{A}$$

Time: $\overline{v'} = 0 \quad v = \overline{v} + v' \quad \overline{\rho \bar{v}_i \bar{v}_j} = \rho \bar{v}_i \bar{v}_j$

$$\overline{\rho v_i v_j} = \frac{1}{t_0}\int_t^{t+t_0} \rho v_i v_j \, dt'$$

Power law: $\tau = k |\dot{\gamma}|^{n-1} \dot{\gamma} \qquad \dot{\gamma} = \frac{\partial v_x}{\partial y}$

BL: $x^* = \frac{x}{L}$; $y^* = \frac{y}{\delta} = \frac{y}{L}\sqrt{Re_L}$; $v_x^* = \frac{v_x}{U}$

$v_y^* = \frac{v_y}{U\delta/L}$; $P^* = \frac{P}{\rho U^2}$; $Re_L = \frac{\rho U L}{\mu}$

$\mu, \rho \rightarrow$ Dynamic viscosity $\quad [\mu] = \frac{M}{LT}$

$Pr[-] \ \frac{ML^2}{T^3}$

$$Re = \frac{\rho v D}{\mu}$$

$$F_D = A_p \cdot C_D \rho \frac{v_0^2}{2} = A_{char} \cdot k E_{char} C_D$$

Math:

$$\tau = \begin{pmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{pmatrix} \qquad a \otimes b = \begin{pmatrix} a_1 b_1 & a_1 b_2 & a_1 b_3 \\ a_2 b_1 & a_2 b_2 & a_2 b_3 \\ a_3 b_1 & a_3 b_2 & a_3 b_3 \end{pmatrix}$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2} \quad ; \quad \sinh(x) = \frac{e^x - e^{-x}}{2} \quad ; \quad \tanh(x) = \frac{\sinh(x)}{\cosh(x)}$$

$\cosh(0) = 1 \qquad \sinh(0) = 0 \qquad \tanh(x) = 0$

$$\cosh^2 x - \sinh^2 x = 1 \qquad \frac{d}{dx}\cosh(x) = \sinh(x)$$

$$\iint_{CS} P \cdot n \, dA = \iiint_{CV} \operatorname{div}(P)\, dV \qquad \underline{\underline{A}} : \underline{\underline{B}} = \sum_i^n A_i B_i$$

Diff eqs:

① $f''(x) + a f'(x) = 0$ → $T(x) - T_\infty = A\sin(\sqrt{a}x) + B\cos(\sqrt{a}x)$ } Complex roots $\alpha = 0$

② $f''(x) - a f'(x) = 0$ → $T(x) - T_\infty = A\sin(\sqrt{a}x) + B\cos(\sqrt{a}x)$

General: $a_1 r^2 + a_1 r + a_2 = 0$ → Solve this to get roots

solvt.

If roots: ① Real $y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$ ② Complex $(\alpha + i\beta)$ $y = e^{\alpha x}(C_1 \cos\beta x + C_2 \sin\beta x)$ ③ Real (Not distinct $r_1 = r_2 = r$) $y = C_1 e^{rx} + C_2 x e^{rx}$

Heat:

$-\int q\cdot n\,dA = hA(T_\infty - T_0)$; $\dot{Q} = -\int q\cdot n\,dA$; $q = -k\nabla T$ (Fourier's Law) → Conduction

$\Downarrow$

$-q\cdot n = h(T_\infty - T_0)$

$V = IR \Rightarrow \Delta T = Q\cdot R$; $S = \frac{\dot{Q}}{k\Delta T}$; $\alpha = \frac{k}{\rho \tilde{c}_p}$

$\dot{Q} = h_{overall}\cdot A_{char}\cdot \Delta T_{char}$

Resistance:

Planar

$\frac{Q}{A} = k\frac{\partial T}{\partial x}$

$Q\,dx = kA\,dT$

$QL = kA\Delta T$

$\Delta T = \dot{Q}\left(\frac{L}{kA}\right)$

Cylindrical

$\dot{Q} = 2\pi r L k \frac{\partial T}{\partial r}$

$\dot{Q}\ln(r_2/r_1) = 2\pi L k \Delta T$

$\Delta T = \dot{Q}\left(\frac{\ln(r_2/r_1)}{2\pi L k}\right)$

Spherical

$Q = -\int q\cdot n\,dA$

$= k\frac{\partial T}{\partial r}(4\pi r^2)$

$\frac{Q}{r^2} = k\frac{dT}{dr}$

$\Delta T = \dot{Q}\left(\frac{r_2 - r_1}{4\pi k r_1 r_2}\right)$

Lumped:

From micro E.

$\rho V \tilde{c}_p \frac{\partial T}{\partial t} = hA(T_\infty - T) + S_c$

$\Downarrow$

For Heat exchanges: $2\pi r$

$m\tilde{c}_p\,dT_c = Ah_h(T_H - T_c)\,dz$

$\hookrightarrow \frac{T_H - T_c}{T_H - T_{sin}} = \exp\left(\frac{-2\pi r_o h}{\dot{m}\tilde{c}_p} z\right)$

$Bi = h\frac{(V/A)}{k}$; $Nu_D = \frac{hD}{k}$; $Pr = \frac{\mu \tilde{c}_p}{k}$; $Gr = \frac{\rho^2 g L^3 \Delta T \beta}{\mu^2}$ $\frac{(Buoyancy)}{(Viscous)}$

$\left(\frac{(Conductive\ (Internal)\ R}{(Convective\ (external)\ R}\right)$ $\left(\frac{(Momentum\ trans.}{Energy\ trans.}\right)$

$Bi \ll 1 \rightarrow$ constant T profile

Initial cond.

$t = 0,\ T(0,y) = T_0$

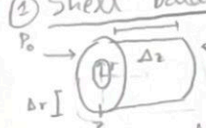
Boundary Cond

$T(L,0) = T_1$ $\quad q_z = 0$

$T(t,S) = T_0$

Ex.:

① Shell balance



$\Sigma F = 0$ (Fully developed)

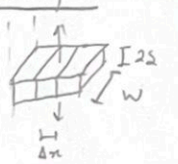
$\Rightarrow P_1A_1 - P_2A_2 + \tau_{rz}A - \tau_{rz}A$

$= P(2\pi r\Delta r)|_z - P(2\pi r\Delta r)|_{z+\Delta z}$

$+ \tau_{rz}(2\pi r\Delta z)|_{r+\Delta r}$

$- \tau_{rz}(2\pi r\Delta z)|_{r-\Delta r}$

② Heat fin



$\rightarrow q_x(2Sw) - q_{x+\Delta x}2Sw - 2h w\Delta x(T(x) - T_\infty) = 0$

$\rightarrow -\frac{dq_x}{dx} - \frac{h}{S}(T(x) - T_\infty) = 0$

$\rightarrow$ Diff eq in terms of T. Express $T(t,y) = \underbrace{T^s(y)}_{} + T^t(t,y)$

$\rightarrow$ Non-dimensional variable: θ

$\rightarrow \theta(t) = f(t)g(t) \Rightarrow$ Separate & solve

$\rightarrow$ Solve diff eq: $\frac{1}{\alpha f(t)}\frac{\partial f(t)}{\partial t} = \frac{1}{g(y)}\frac{\partial^2 g(y)}{\partial y^2} = \pm\lambda^2$ for one value for λ

$\rightarrow$ Solve for constants (sin & cos will have series)

$\theta = \sum_{n=-\infty}^{\infty} \exp(-\lambda_n^2 \alpha t)\,\tilde{A}_n \cos(\lambda_n y)$ $\quad \lambda_n = \left(\frac{n\pi + \frac{\pi}{2}}{b}\right)$

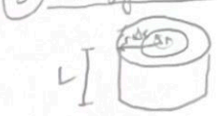
$\hookrightarrow$ Use orthogonality to solve for $\tilde{A}_n$ & other constants

Orthogonality:

$\left.\begin{aligned}\int_0^b \sum \tilde{A}_n \cos(\lambda_n y)\cos(\lambda_m y)\,dy \\ \int_0^b \sum \tilde{A}_n \sin(\lambda_n y)\sin(\lambda_m y)\,dy\end{aligned}\right\} = 0 \quad (n \neq m)$

$\left.\begin{aligned}\\ \\ \end{aligned}\right\} = K \quad (m = n)$

③ Energy Shell balance.

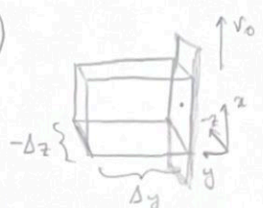


$Acc = In - Out + gen.$

$q_r(2\pi rL)|_r - q(2\pi rL)|_{r+\Delta r} + S_c 2\pi r\Delta r L$

$\frac{\partial(rq_r)}{\partial r} = S_c r$

④



$\Sigma F = +\tau_{yx}\Delta z\Delta x - \tau_{yx}\Delta z\Delta x$

$- \rho(V) g_z$

$\Delta x\Delta y\Delta z$

Double angle:

$\cos^2 x = \frac{\cos 2x + 1}{2}$ $\quad \sin^2 x = \frac{1 - \cos 2x}{2}$

Cheat Sheet

$-\frac{dP(z)}{dz} - \rho g = 0 \leftarrow$ [Barometric eq.] $\longrightarrow \nabla P + \rho g = 0$ (Vector form)

$P(z_2) - P(z_1) = -\rho g (z_2 - z_1)$

Flow

$\tau_{yx} = \mu \frac{\partial v_x}{\partial y} = \frac{1}{A_y} \frac{\partial (m v_x)}{\partial t} = \mu \dot{\gamma}$

$\dot{\gamma}_{yx} = \frac{\partial v_x}{\partial y}$ (Strain rate)

$t_{char} = \frac{\rho h^2}{\mu}$

$[\mu]$ = poise = $\frac{g}{cm \cdot s}$

$\frac{\mu}{\rho} \rightarrow$ kinematic viscosity

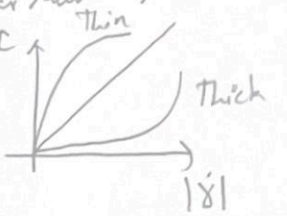
$\mu = T^{1/2}$ (Ideal gas)

$\mu = e^{a/T}$ (Liquids)

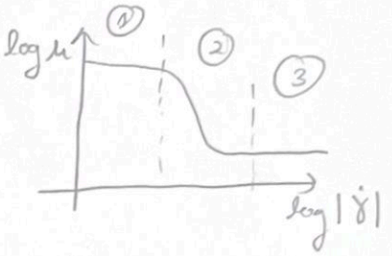
Power = $F \cdot v$
$= A \tau_{yx} v$
$= A \mu \frac{\partial v}{\partial y} \cdot v$
$= \frac{A \mu v^2}{h}$

Newtonian [Power law & μ is constant]

$\tau = K |\dot{\gamma}|^{n-1}$



① $n = 1$
② $n < 1$
③ $n > 1$



$\langle v \rangle = \frac{1}{A} \iint_{cv} v \cdot n_\perp dA$

$dA = r\, dr\, d\theta = r^2 \sin\phi\, d\cdot d\phi\, d\theta$ (↳ n, k, y)

$\frac{dm}{dt} = \dot{m}_{in} - \dot{m}_{out}$

Balances

Macroscopic Mass (integral)

$\frac{\partial}{\partial t} \iiint_{cv} \rho\, dV = - \iint_{cs} v_{in} \cdot n_\perp dA - \iint_{cs} v_{ar} \cdot n_\perp dA$

Microscopic Mass balance (continuity eq)

$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \underline{v}) = 0$

(Also used for compressibility equation)

Macroscopic momentum balance

$\Sigma F = \frac{\partial}{\partial t} \iiint_{cv} \rho v \cdot dV + \rho \iint v \cdot (v \cdot n)_\perp dA$

↳ (Using average vel. & β)

$\Sigma F = \frac{\partial}{\partial t} \iiint_{cv} \rho \underline{v}\, dV + \sum_{cs} \rho \langle v^2 \rangle A\, n_\perp$

$\Sigma F = \frac{\partial}{\partial t} \iiint_{cv} \rho \underline{v}\, dV + \sum_{cs} \dot{m} \langle v \rangle \beta\, n_\perp$

$\beta = \frac{\langle v^2 \rangle}{\langle v \rangle^2}$

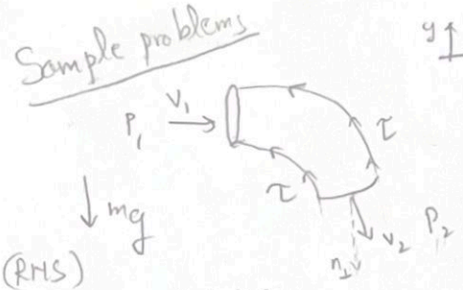
Cauchy (Micro momentum) or Force balance

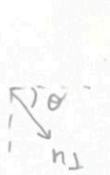
x: $\frac{\partial}{\partial t}(\rho v_x) + \frac{\partial}{\partial x}(v_x \rho v_x) + \frac{\partial}{\partial y}(v_y \rho v_x) + \frac{\partial}{\partial z}(v_z \rho v_x)$
$= -\frac{\partial P}{\partial x} + \frac{\partial}{\partial x}(\tau_{xx}) + \frac{\partial}{\partial y}(\tau_{yx}) + \frac{\partial}{\partial z}(\tau_{zx}) + \rho g_x$

y: $\frac{\partial}{\partial t}(\rho v_y) + \frac{\partial}{\partial x}(v_y \rho v_x) + \frac{\partial}{\partial y}(v_y \rho v_y) + \frac{\partial}{\partial z}(v_y \rho v_z)$
$= -\frac{\partial P}{\partial y} + \frac{\partial}{\partial x}(\tau_{xy}) + \frac{\partial}{\partial y}(\tau_{yy}) + \frac{\partial}{\partial z}(\tau_{zy}) + \rho g_y$

z: Similar ($\tau_{xz}, \tau_{yz}, \tau_{zz}$)

Sample problems





(RHS)

$$\Sigma F = \sum_{CS} \dot{m}\,\langle v\rangle\,\beta\, n_\perp$$

$$= \dot{m}_1 \langle v_1\rangle \beta_1 (-e_x) + \dots$$

$$+ \dot{m}_2 \langle v_2\rangle \beta_2 (\cos\theta\, e_x - \sin\theta\, e_y)$$

(LHS)

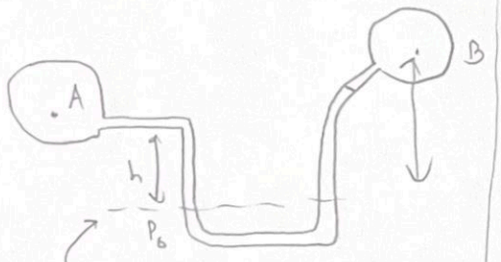
$$\Sigma F = P_1 A_1 e_x - P_2 A_2 (\cos\theta\, e_x - \sin\theta\, e_y) - m_F g\, e_y \underbrace{- \iint_{CS} \rho\, n_\perp\, dA + \iint_{CS} \underline{\tau}\, n_\perp\, dA}_{F_{wf}}$$

Resolving:

$$x: \quad (F_{fw})_x = P_1 A_1 - P_2 A_2 \cos\theta - \dot{m}_1 \langle v_1\rangle \beta_1 + \dot{m}_2 \langle v_2\rangle \beta_2 \cos\theta$$

$$y: \quad F_{fw} = + P_2 A_2 \sin\theta - m_F g - \dot{m}_2 \langle v_2\rangle \beta_2 \sin\theta$$

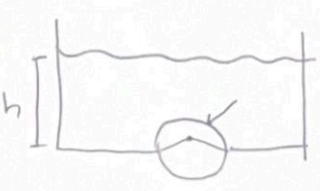
(Only dependent on v_2 and as $v_2 \uparrow$, F_y increases)



+ Always have a reference
+ Make sure the correct densities of two liquids are taken (they may be different)

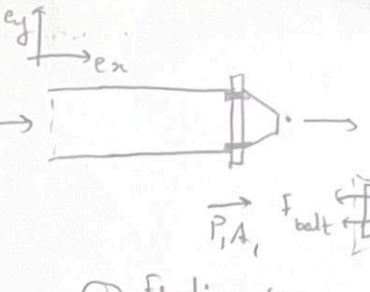
$$(P_A - P_b) = -\rho g h$$

$$(-)\ (P_B - P_b) = -\rho g d,$$



$$h(\theta) = h - r\cos\alpha + r\cos\theta$$

$$F_y = \underbrace{(\rho g h(\theta))}_{P} \cdot A$$



Forces on bolt

① Finding v_2:

$$\rho \langle v_1\rangle A_1 = \rho \langle v_2\rangle A_2$$

$$\langle v_1\rangle = \frac{\langle v_2\rangle A_2}{A_1}$$

② $\Sigma F = \frac{\partial}{\partial t}\int_{CV} \rho\, \underline{v}\, dV + \sum_{CS} \beta \dot{m} \langle v\rangle n_\perp$

$$\rho \langle v\rangle A$$

$$\Sigma F = \rho V_1^2 A_1 (-e_x) + \rho V_2^2 A_2 (e_x)$$

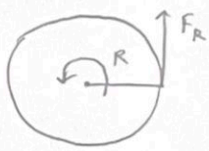
$$\Sigma F_x = (F_B)_x + P_1 A_1 - P_2 A_2 - P_a (A_3 + A_4)$$

$$= (F_B)_x + P_1 A_1 - P_a (A_2 + A_3 + A_4)$$



$$(F_B)_x + (P_1 - P_a) A_1 = \rho V_1^2 A + \rho V_2^2 A_2$$

$$(P_a)_x = P_a \cos\theta$$

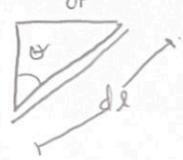


Torque (i)

$$i = r \times F_R$$

$$= r F_r$$

$$\int_0^{D/2} r F_r$$



$$dl = \frac{dr}{\sin\theta}$$

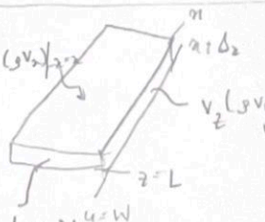
$$dF_R = \tau \cdot dA$$

$$= \mu \frac{(v_2 - v_1)}{h} dA$$

$$dF_R = \frac{\mu\, \omega\, r\, dA}{h}$$

→ F_R exerted on a slide







→ This is the normal term

$$F_z = \left[v_z \rho V_z|_{z=L}\right] W \Delta x$$

(6 terms)

Force balance:
LW, $W\Delta x$
$L\Delta x$
(6 terms + gravity)
Δ pressure

$$\left[\tau_{xz}|_{x+\Delta x} - \tau_{xz}|_x\right] WL$$